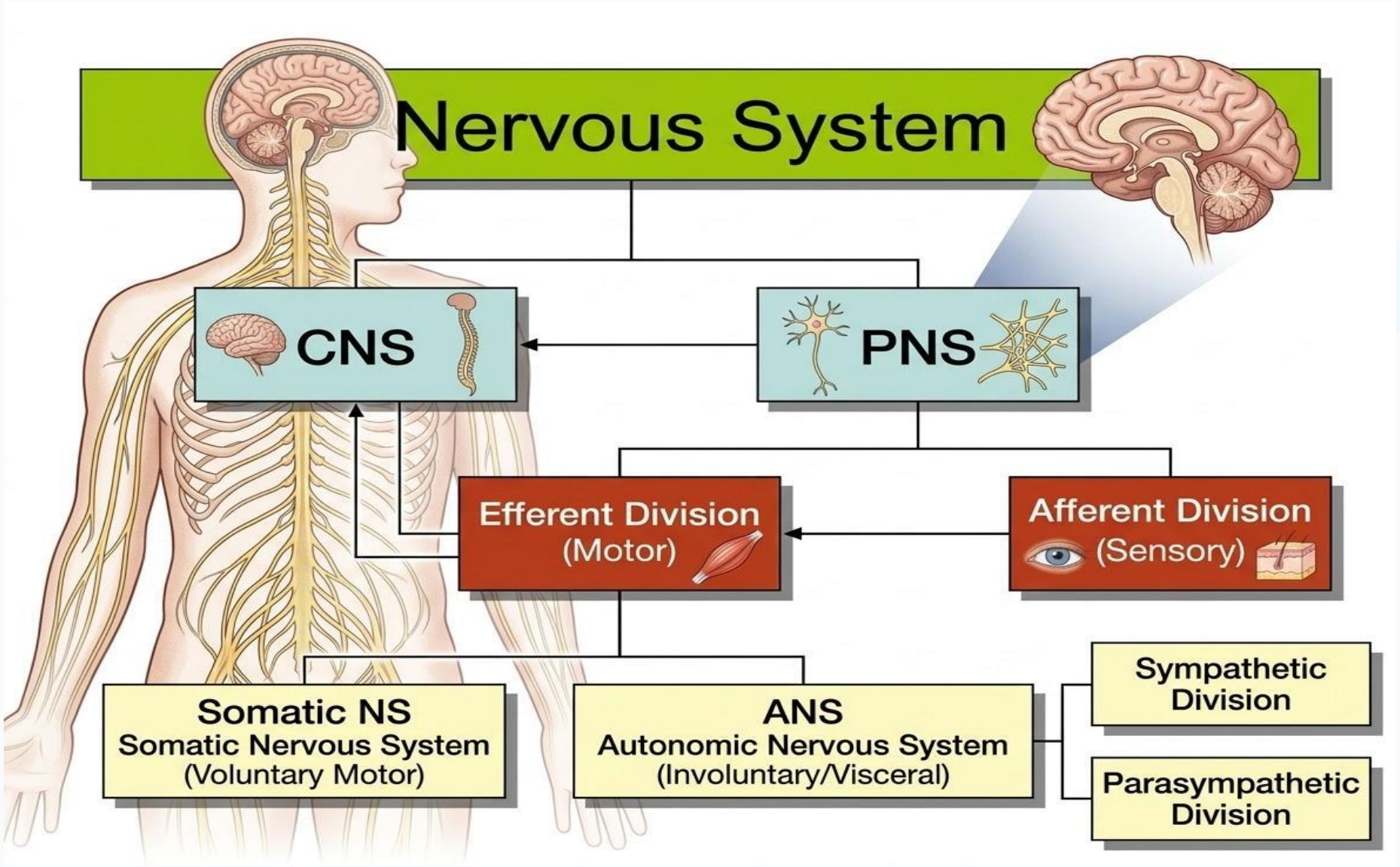
Pharmacology Lecture Series

Drugs Affecting the Autonomic Nervous System

An Introduction to ANS Physiology & Neurotransmission

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Nervous System Hierarchy



Nervous System Hierarchy

Central NS (CNS)

Consists of the brain and spinal cord. Serves as the primary control center for processing information and issuing commands.

Peripheral NS (PNS)

Connects the CNS to limbs and organs. Divided into **Afferent** (Sensory input) and **Efferent** (Motor output) pathways.

Efferent Divisions

Comprises the **Somatic NS** (voluntary skeletal control) and the **Autonomic NS** (involuntary visceral regulation).

Somatic vs. Autonomic System

Somatic Nervous System

Voluntary Control: Regulates functions under conscious mind control.
Effector: Direct innervation of skeletal muscles to drive locomotion and somatic reflexes.

Autonomic Nervous System

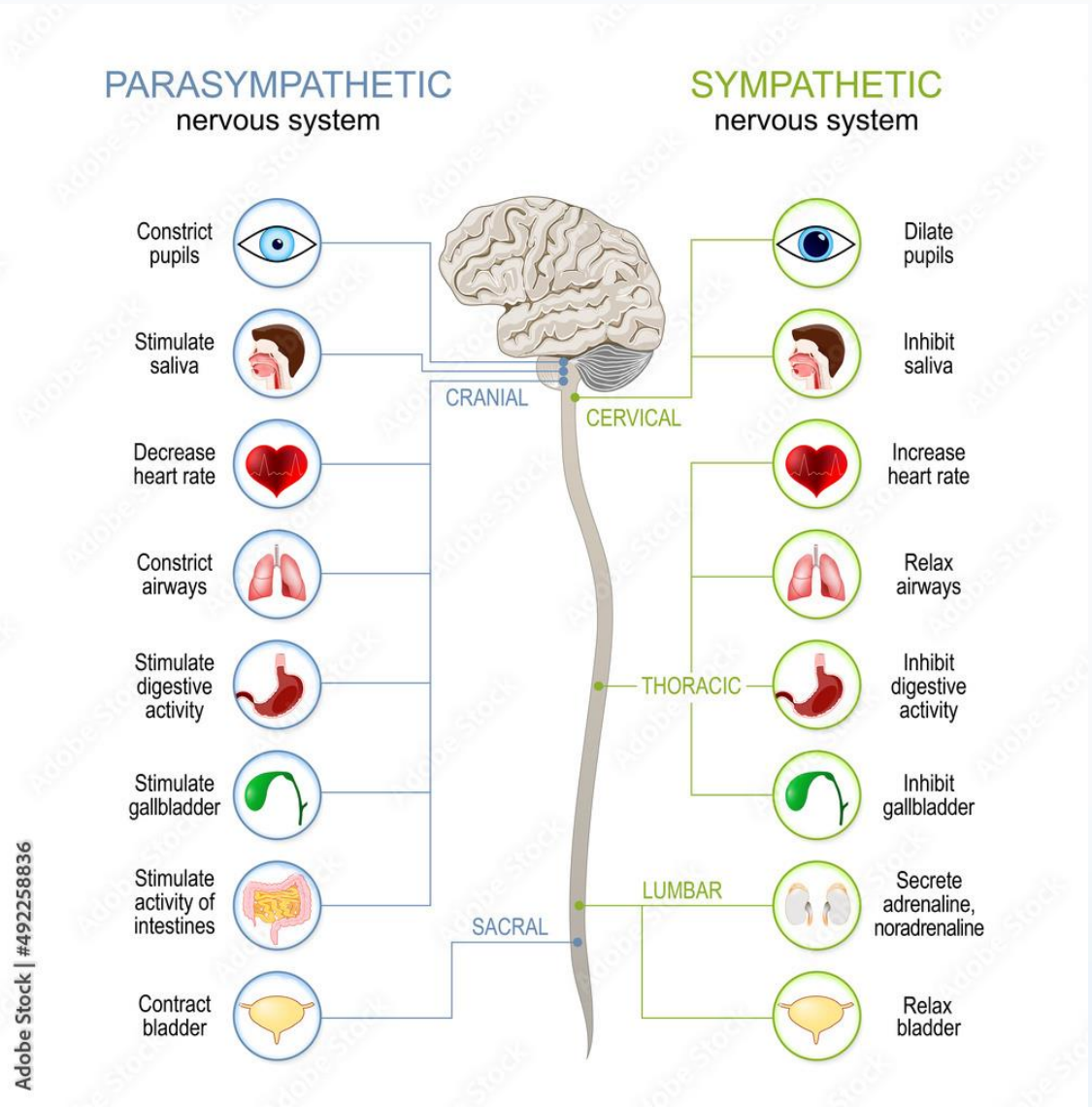
Involuntary Control: functions involuntarily to regulate the everyday needs and requirements of the body without the conscious participation of the mind.

Effector: Regulates cardiac muscle, smooth muscle, and glandular visceral functions (e.g., HR, BP).

ANS Efferent Neuronal Pathways

The Autonomic Nervous System transmits nerve impulses from the CNS to effector organs via two distinct efferent neurons in series:

- **Preganglionic Neurons:** Cell body originates within the CNS (brainstem or spinal cord); its myelinated axon extends to an autonomic ganglion.
- **Autonomic Ganglia:** Synaptic relay station where preganglionic neurons release neurotransmitters to stimulate postganglionic cells.
- **Postganglionic Neurons:** Unmyelinated axon originates in the ganglion and projects directly to effector tissues (cardiac/smooth muscle, glands).



Mechanism of Neurotransmission

1. Conduction

The propagation of an electrical action potential along the axonal nerve membrane from the cell body down to the nerve terminal.

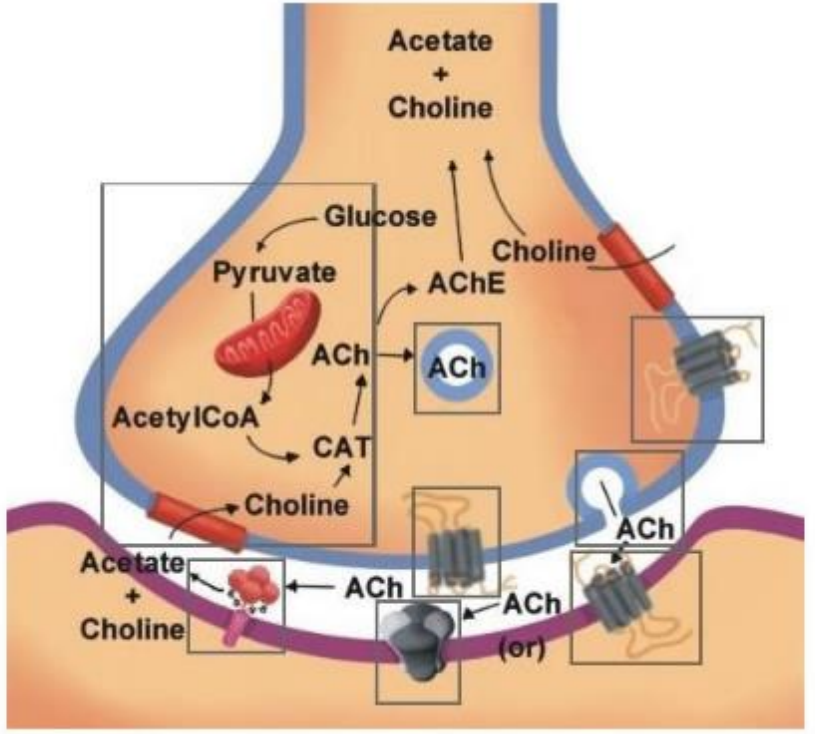
2. Synaptic Transmission

The chemical transfer of nerve impulses across ganglion synapses and neuroeffector junctions mediated by specific chemical messengers (neurotransmitters).

Acetylcholine Function Sites

Acetylcholine (ACh) serves as the key neurotransmitter at multiple neuroeffector and synaptic junctions:

- ✓ **All Preganglionic Nerve Endings:** Including both sympathetic and parasympathetic ganglia, as well as the adrenal medulla.
- ✓ **Postganglionic Parasympathetic:** All parasympathetic effector sites.
- ✓ **Specific Sympathetic Sites:** Postganglionic sympathetic fibers supplying sweat glands and skeletal muscle vasodilator vessels.
- ✓ **Somatic & CNS:** Present at Neuromuscular Junctions (NMJ) and within the brain.



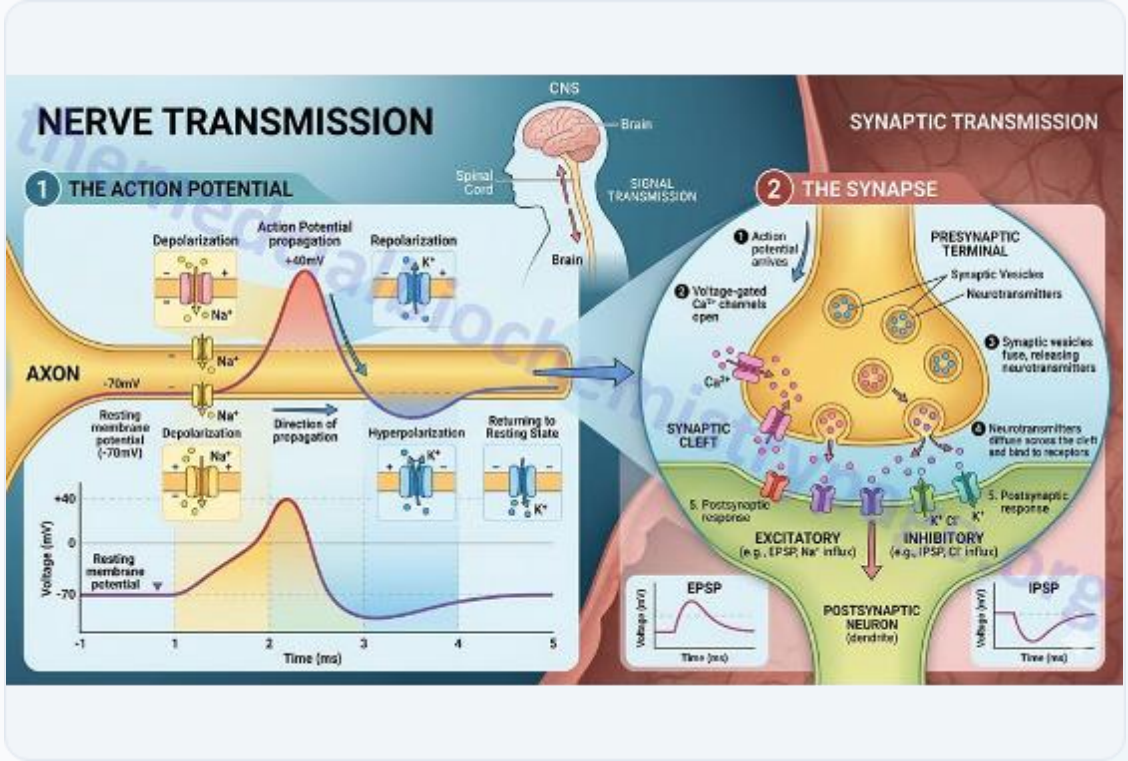
Noradrenaline & Neuronal Types

Noradrenaline (Norepinephrine)

Primary neurotransmitter at almost all postganglionic sympathetic nerve endings (**except** sweat glands & skeletal muscle vasodilator vessels).

Neuronal Classification

- **Cholinergic Neurons:** Neurons that synthesize, store, and release *Acetylcholine* as their primary transmitter.
- **Adrenergic Neurons:** Neurons that synthesize, store, and release *Noradrenaline* as their chemical messenger.



Acetylcholine Receptors

1. Muscarinic Receptors

Receptor Class: G-protein coupled transmembrane receptors (GPCRs).
Primary Locations: Parasympathetic neuroeffector sites (heart, smooth muscle, glands), autonomic ganglia, and central nervous system.
Response: Mediates slow metabolic responses via secondary messenger cascades.

2. Nicotinic Receptors

Receptor Class: Ligand-gated ion channels (ionotropic receptors).
Primary Locations: Autonomic ganglia, adrenal medulla, and skeletal muscle neuromuscular junction (NMJ).
Response: Mediates rapid membrane depolarization via ion influx.

Nicotinic Receptors & Blockers

- **Receptor Distribution:** Found in autonomic ganglia, adrenal medulla chromaffin cells, and skeletal muscle motor endplates.
- **Subtype Heterogeneity:** Nicotinic receptors in autonomic ganglia vs. skeletal muscle are structurally non-identical. Both are activated by acetylcholine but possess distinct pharmacological sensitivity to antagonists.
- **Hexamethonium:** Acts as a selective nicotinic receptor antagonist at autonomic ganglia (*Ganglion Blocker*).
- **d-Tubocurarine:** Acts as a selective nicotinic receptor blocker at the neuromuscular junction in skeletal muscle.

Adrenergic Receptors

Receptor Subtype	Selective Agonist	Selective Antagonist
α1 (Alpha-1)	Phenylephrine	Prazosin
α2 (Alpha-2)	Clonidine	Yohimbine
β1 (Beta-1)	Denopamine	Metoprolol
β2 (Beta-2)	Salbutamol	Butoxamine
β3 (Beta-3)	Mirabegron	SR 59230A

Muscarinic Receptors

Receptor Subtype	Selective Agonist	Selective Antagonist
M1	Oxotremorine	Telenzepine
M2	Methacholine	Methoctramine
M3	Bethanechol	Oxybutynin
M4		
M5		

** Note: Hexamethonium is classified specifically as a ganglionic nicotinic blocker.*

Questions & Discussion

Thank you for your attention

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